Tinnaphop Sonsang

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PROFESSIONAL SUMMARY

First-Class Honors Computer Engineering & AI graduate with hands-on experience designing end-to-end machine learning, predictive modeling, and computer vision solutions. Skilled in Python, SQL, statistical analysis, data cleansing, feature engineering, and automated data workflows. Proven track record in execution of experiments, model evaluation, and deploying AI/ML pipelines to support data-driven decision making.

EDUCATION

Walailak University — Nakhon Si Thammarat
B.Eng. Computer Engineering and Artificial Intelligence — June 2022 – 2026
GPAX: 3.65 (First-class honors) | Graduation: 7 May 2026

CERTIFICATIONS

EF SET English Certificate (View certificate: cert.efset.org/en/gCMcLJ) — *Awarded 3 Aug 2026*

- Overall score: 56/100 — B2 Upper Intermediate (CEFR); Reading: C1 Advanced (64/100)

KEY SKILLS

- **Data Gathering, Cleansing & EDA:** Python (Pandas, NumPy), SQL, Data Cleansing, Data Ingestion, Exploratory Data Analysis (EDA), Knowledge of Statistics, Distribution Analysis, Outlier Detection.
- **Data & Feature Engineering:** Feature Engineering, Feature Selection, Dimensionality Reduction (PCA, UMAP), Feature Transformation (Yeo-Johnson), Unsupervised Dataset Deduplication (KMeans, GMM).
- **Machine Learning & Predictive Analytics:** Machine Learning, Predictive Modeling, Execution of Experiments, Computer Vision (YOLO, RT-DETR, Grounding DINO), Audio Signal Processing, Neural Networks (Complex-Valued NNs, U-Net), Learn-to-Rank.
- **Model Validation & Evaluation:** Precision, Recall, F1-Score, Confusion Matrix, mAP, SI-SDR, Silhouette Score, Model Validation & Performance Testing.
- **Deployment, Cloud & Data Visualisation:** Microsoft Azure (Azure ML Compute via SSH), RESTful APIs, Cloud GPU Infrastructure, Data Visualisation (Matplotlib, Seaborn, Plotly.js), Interactive Dashboards, Git, Docker.
- **Full Stack Web Development:** Next.js, React, TypeScript, Astro, Interactive Analytics Dashboards, Full-Stack Integration.

WORK EXPERIENCE

Intern Developer (Co-op/Internship) | ThaiDotRun (merged into ASICS), Bangkok — Sept 2025 – Apr 2026

Automated Runner Highlight Extraction System

- Evaluated multiple object detection architectures (including YOLO and RT-DETR), selecting RT-DETR for its Detection Transformer architecture's superior global context modeling in crowded race scenes, paired with BoTSORT for robust subject tracking.
- Implemented a custom spatial weighting algorithm prioritizing camera-center subjects and integrated OCR bib number recognition to dynamically replace track IDs for frame-accurate highlight extraction.
- Achieved an average success rate of approximately 60% in automated highlight video generation, establishing a solid baseline for initial production deployment and future pipeline optimization.

Robustness Shoe Detection & Classification Pipeline

- Designed and developed an end-to-end deep learning pipeline to detect and classify shoe brands from complex race images and video footage.
- Applied unsupervised learning for dataset deduplication to eliminate redundant training frames, using Grounding DINO to localize runners and extract shoe crops for ensemble model training.
- Integrated depth-mapping and keypoint detection to assess image quality and enhance overall prediction reliability.
- Successfully deployed an ensemble classification framework that achieved strong performance on major brands (Adidas, Nike, ASICS), while establishing automated quality control and data re-collection workflows to improve recognition for long-tail brands.

Data Scientist | Lief Capital Asset Management, Bangkok Short Term: May 2026

- Analyzed and refactored an existing financial data and machine learning codebase on Azure ML, restructuring monolithic workflows across data gathering, preprocessing, feature engineering, model training, and evaluation to improve maintainability and scalable experimentation.
- Optimized the end-to-end data processing and EODHD ingestion pipeline, reducing runtime from over 1 hour to under 30 minutes by adopting Polars and scaling parallel network workers.
- Worked on an XGBoost-based stock ranking pipeline, including sector-relative feature engineering, NDCG performance evaluation, hyperparameter tuning with Optuna, and quarterly backtesting.

PROJECTS

Complex-Valued Neural Networks: When Does Phase Actually Help? *July 2026 – Present*

- Investigated whether complex-valued neural networks can outperform real-valued models by identifying where in the learning pipeline complex representations help or hurt performance.
- Built complex-valued CNNs, U-Nets, and a 2-source audio separation benchmark from scratch, confirming that phase information carries meaningful signal (~33 dB drop when removed).
- Ran a 60-run factorial experiment (15 seeds per configuration) with statistical analysis, finding that performance depends on the interaction between activation function and loss function, not on either factor alone.
- Traced the interaction through forward geometry, gradient propagation, and parameter updates, showing that the optimizer mediates how architectural choices translate into learning dynamics and final performance.

Madonna Discography Analysis (Sonic Atlas — site(.)moss) *March 2026 – August 2026*

- Built an end-to-end unsupervised machine learning and data analytics pipeline using Spotify audio-feature data, including exploratory data analysis (EDA), data preprocessing, deduplication, feature engineering, and correlation analysis.
- Applied Yeo-Johnson power transformation for feature normalization and distribution correction, preventing outlier tracks from forming non-meaningful micro-clusters.
- Benchmarked KMeans clustering and Gaussian Mixture Models (GMM) at K=4 using Silhouette Score and Davies-Bouldin Index, and compared GMM covariance structures using BIC/AIC model selection.
- Designed a statistical validation framework using bootstrap resampling, 50-seed stability analysis, feature-subset sensitivity testing, and per-sample silhouette distributions to evaluate clustering robustness and reproducibility.
- Applied UMAP dimensionality reduction with topology-sensitivity analysis and automatically generated interpretable cluster labels from per-cluster z-scores.

Sitedotmoss (site(.)moss) *December 2025 – Present*

- Developed and deployed a full-stack web application using Astro (SSR), React 19, TypeScript, and Turso/LibSQL, featuring a responsive dashboard, blog CRUD system, and integrated AI/ML inference demos.
- Designed a normalized SQLite relational database schema with 5 tables and composite indexes using Drizzle ORM, implementing 8 RESTful API endpoints for full CRUD operations.
- Implemented authentication and session management using PBKDF2 password hashing and secure session handling.
- Integrated real-time YOLOv12 object detection inference from the Capstone Project through Hugging Face Spaces and Gradio REST API, enabling AI model deployment within the web application.
- Engineered modular React components with optimistic UI updates and built an interactive data visualization dashboard using Plotly.js, alongside custom CSS/canvas animations.

Capstone Project (Thesis) — Detection of White Blood Cells in Raptors *August 2024 – August 2025*

- Developed deep learning and computer vision object detection models for automated detection of six blood-cell classes in raptor blood smear images using 1,180 manually annotated images.
- Benchmarked YOLO-11l, YOLO-12l, YOLO-26l, and RTDETR-l across five data augmentation strategies, evaluating model performance using mAP, precision, and recall. Achieved a best baseline performance of mAP@0.5 = 0.957 and mAP@0.5:0.95 = 0.746, with 93.4% recall, using YOLO-26l.
- Evaluated Mosaic, MixUp, CLAHE, photometric, and geometric data augmentation techniques and applied Grad-CAM for Explainable AI (XAI) and model interpretability.
- Built and deployed a full-stack AI inference application using Next.js, React, and PostgreSQL (Supabase) for batch image uploads and automated object detection results
- Manuscript, “YOLO-Based Detection in Wildlife Hematology: Enhancing White Blood Cell Recognition with Data Augmentation,” submitted to *Computer Methods and Programs in Biomedicine* (CMPB-D-26-04827).